

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XVII

JEE (Main)-2022

TEST DATE: 14-07-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

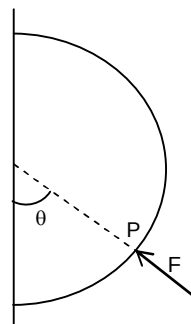
Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A hollow uniform hemisphere of radius R is pressed against a smooth vertical wall by a force F at point P as shown. The force F makes an angle θ with vertical and its line of action is passing through center of hemisphere. Center of mass of hemisphere is at distance $\frac{R}{2}$ from the center. What is the value of $\tan \theta$ for equilibrium?

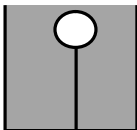


- (A) $\frac{1}{\sqrt{3}}$
 (B) $\sqrt{3}$
 (C) 2
 (D) $\frac{1}{2}$
2. A long cylinder has radius R and charge is distributed symmetrically in its volume such that electric field at distance r ($r < R$) from the axis is E . What is charge density at distance r ($r < R$)?
- (A) $\rho = \frac{1}{\epsilon_0} \left[\frac{dE}{dr} \right]$
 (B) $\rho = \epsilon_0 \left[\frac{dE}{dr} \right]$
 (C) $\rho = \frac{1}{\epsilon_0} \left[\frac{dE}{dr} + Er \right]$
 (D) $\rho = \epsilon_0 \left[\frac{dE}{dr} + \frac{E}{r} \right]$
3. Two moving coil galvanometers X and Y are connected in series and a current passes through them. Their readings are found to be full scale for X and half scale for Y. Given that their scales are identical and their restoring springs of same strength, but that X's coil is twice as large in area and has twice the number of turns compared with Y's coils. The magnetic field strength of X's magnet compared with Y's magnet is :-
- (A) $\frac{1}{4}$
 (B) $\frac{1}{2}$
 (C) 1
 (D) 4

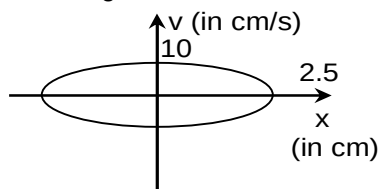
4. The electric field of a radio wave is given by $\vec{E} = E_0 \sin(kz - \omega t)(\hat{i} + \hat{j})$. What is the unit vector in the direction of the magnetic field at a place and time where $\sin(kz - \omega t)$ is positive?

- (A) $\frac{(\hat{i} + \hat{k})}{\sqrt{2}}$
 (B) $\frac{(-\hat{i} - \hat{k})}{\sqrt{2}}$
 (C) $\frac{(-\hat{i} + \hat{j})}{\sqrt{2}}$
 (D) $\frac{(\hat{i} - \hat{j})}{\sqrt{2}}$

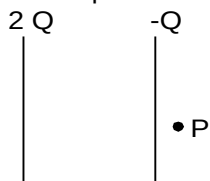
5. A cork suspended from the bottom of a container filled with water with a string as shown in figure. If the container accelerates in a horizontal direction towards right, which one is correct?



- (A) Inclination of buoyant force with vertical is $\tan^{-1}(a/g)$ towards left
 (B) Inclination of the buoyant force with vertical is $\tan^{-1}(a/g)$ towards right
 (C) Inclination of the buoyant force with vertical is $[\pi/2 - \tan^{-1}(a/g)]$ towards left
 (D) Inclination of the buoyant force with vertical is $[\pi/2 - \tan^{-1}(a/g)]$ towards right
6. Assume that a particle cannot be confined to a spherical volume of diameter less than de-Broglie wavelength of the particle. Estimate the minimum kinetic energy a proton confined to a nucleus of diameter 10^{-14} m may have. (Mass of proton = 1.67×10^{-27} kg)
- (A) 10.6 MeV
 (B) 12 MeV
 (C) 8.2 MeV
 (D) 21 MeV
7. ^{32}P beta-decays to ^{32}S . Find the sum of the energy of the antineutrino and the kinetic energy of the β -particle. Neglect the recoil of the daughter nucleus. Atomic mass of $^{32}\text{P} = 31.974$ u and that of $^{32}\text{S} = 31.972$ u. ($1 \text{ u} = 931 \text{ MeV}/c^2$)
- (A) 1.2 MeV
 (B) 1.862 MeV
 (C) 1.243 MeV
 (D) 2.162 MeV
8. The figure shows a graph between velocity and displacement (from mean position) of a particle performing SHM. select the incorrect statement

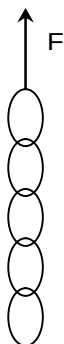


- (A) the time period of the particle is 2s
 (B) the maximum acceleration will be 40cm/s^2
 (C) the velocity of particle is $2\sqrt{21}\text{ cm/s}$ when it is at a distance 1 cm from the mean position.
 (D) Amplitude of SHM is 2.5 cm.
9. A Carnot engine uses first an ideal monatomic gas ($\gamma = 5/3$) and then an ideal diatomic gas ($\gamma = 7/5$) as its working substance. The source and sink temperature are 411°C and 69°C respectively and the engine extracts 1000 J of heat from the source in each cycle. Then :
 (A) the efficiencies of the engine in the two cases are in the ratio 21 : 25
 (B) the area enclosed by the P-V diagram in the first case is 600 J
 (C) the area enclosed by the P-V diagram in both cases is 500 J
 (D) the heat energy rejected by the engine in the first case is 600 J while that in the second case is 714.3 J
10. In the figure shown the plates of a parallel plate capacitor have unequal charges. Its capacitance is 'C'. P is a point outside the capacitor and close to the plate of charge $-Q$. The distance between the plates is 'd'. Select incorrect alternative.



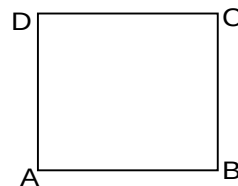
- (A) A point charge at point 'P' will experience electric force due to capacitor
 (B) The potential difference between the plates will be $3Q/2C$
 (C) The energy stored in the electric field in the region between the plates is $\frac{9Q^2}{8C}$
 (D) The force on one plate due to the other plate is $\frac{Q^2}{2\pi\epsilon_0 d^2}$

11. A chain consists of 5 links each of mass m . First link is being pulled up by a force F . What is force between 2nd and 3rd link?



- (A) $\frac{3F}{5}$, only if $F > 5mg$
 (B) $\frac{2F}{5}$, only if $F < 5mg$
 (C) $\frac{2F}{5}$, only if $F = 5mg$
 (D) $\frac{3F}{5}$ in all cases.

12. Four persons having masses 10 kg, 20 kg, 30 kg and 40 kg are standing at the corners A, B, C & D respectively of square plank of side 2 m and mass 60 kg. There is no friction between plank and the horizontal surface below it. What is magnitude of displacement of plank when all person meet at the center of plank?



- (A) 5 cm
(B) 15 cm
(C) 25 cm
(D) 50 cm

13. An inductor of inductance L is connected across a capacitor of capacitance C . Inductor is kept in a magnetic field such that flux through it is ϕ . Magnetic field is switched off at $t = 0$. What will be maximum charge on capacitor?

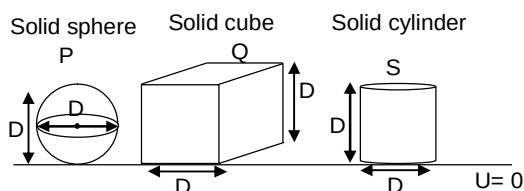
- (A) $\frac{\phi}{2} \sqrt{\frac{1}{LC}}$
(B) $\phi \sqrt{\frac{C}{L}}$
(C) $\frac{1}{\phi \sqrt{LC}}$
(D) $\phi \sqrt{\frac{L}{C}}$

14. Assume gravitational potential energy 'U' at ground level to be zero. All objects are made up of same material.

U_P = Gravitational potential energy of solid sphere

U_Q = Gravitational potential energy of solid cube

U_S = Gravitational potential energy of solid cylinder

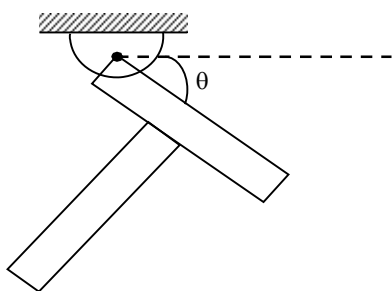


- (A) $U_Q > U_S > U_P$
(B) $U_S > U_Q > U_P$
(C) $U_P > U_Q > U_S$
(D) $U_Q > U_P > U_S$

15. The magnetic susceptibility of a paramagnetic substance at -73°C is 0.0040, then its value at -173°C will be :

- (A) 0.0030
(B) 0.0120
(C) 0.0180
(D) 0.0080

16. A simple telescope consisting of an objective of focal length 30 cm and a single eye lens of focal length 5 cm is focused on a distant object in such a way that parallel rays emerge from the eye lens. If the object subtend an angle of 2° at the objective then the angular width of the image is
 (A) 10°
 (B) 12°
 (C) 24°
 (D) 36°
17. In meter bridge experiment, an error may arise due to least count of meter scale. This error can be minimized by adjusting resistance of resistance box such that null point is obtained near
 (A) 100 cm mark
 (B) 75 cm mark
 (C) 50 cm mark
 (D) 25 cm mark
18. A ray of light travels from medium 2 to medium 1. Relative permittivity and relative permeability of two media are $K_2 = 8$, $K_1 = 6$, $M_2 = 1.5$, $M_1 = 4$ respectively. What is critical angle of incidence in this case?
 (A) 30°
 (B) 45°
 (C) 60°
 (D) 37°
19. An ideal monatomic gas undergoes a process in which its pressure is directly proportional to the square of absolute temperature. By increasing the volume from $V_0 = 2$ litre to $3V_0$, internal energy decreases by 300 J. What is the gas pressure at volume V_0 ?
 (A) $0.9 \times 10^5 \text{ Pa}$
 (B) 10^5 Pa
 (C) $1.5 \times 10^5 \text{ Pa}$
 (D) $2 \times 10^5 \text{ Pa}$
20. A T shaped rod is hinged at its corner as shown and free to rotate in a vertical plane. The rods have mass m each and length λ each. Find angle θ made by upper end in the equilibrium position.

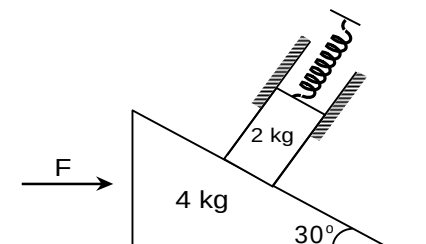


- (A) $\tan^{-1}\left(\frac{1}{2}\right)$
 (B) $\tan^{-1}(2)$
 (C) 37°
 (D) 53°

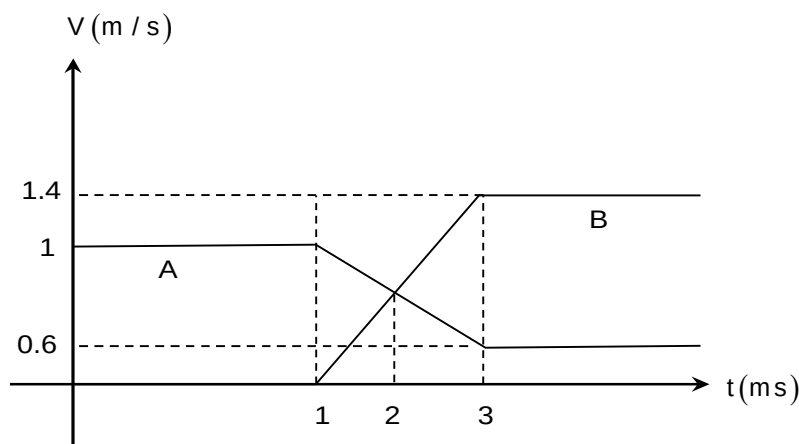
SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A block of mass 2 kg is connected to a spring ($K = 25 \text{ N/m}^2$) and kept on a triangular wedge of mass 4 Kg as shown. Block is constrained to move only along normal to inclined plane of wedge. Find work (in J) required to slowly move the wedge by 2 m to the right. Assume block remain in contact with incline plane, neglect friction & $g = 10 \text{ m/s}^2$.



22. Two sound waves of slightly different frequencies have amplitude ratio $\frac{11}{9}$. What is the difference of sound levels in decibels of maximum and minimum intensities heard at a point :-
23. A ring of radius R has charge $-Q$ distributed uniformly over it. If the charge (q) should be placed at the center of the ring such that the electric field becomes zero at a point on the axis of the ring distance R from the center of the ring, where the value of q is $(Q/4)\sqrt{x}$. Find x.
24. The poisson's ratio of a material is 0.4. If a force is applied to a wire of this material, there is a decrease of cross-sectional area by 2%. The percentage increase in its length is
25. A galvanometer of resistance 7.5Ω contains 100 division. It gives a deflection of one division on passing a current of 10^{-4} A . Find the resistance in ohms to be connected to it, so that it becomes a voltmeter of range 10V.
26. Two blocks A and B are moving along x-axis and collides. Velocity time graph of two blocks are as shown:



What is coefficient of restitution?

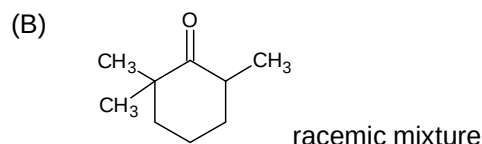
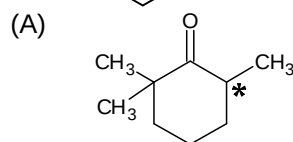
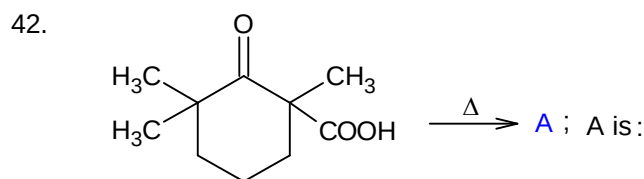
27. In a series LCR ac circuit, power dissipated is half of maximum power when angular frequency of source is 25 rad/s or 64 rad/s. What is resonant angular frequency (in rad/s) of circuit?
28. A vernier scale has certain negative zero error. Its main scale division is 1 mm and 10 vernier scale division is equal to 9 main scale division. When inner diameter of a hollow cylinder is measured then 4th VSD coincides with 3.2 cm mark on main scale and when outer diameter is measure then 8th VSD coincides with 5.6 cm mark on main scale. What is thickness (in cm) of wall of hollow cylinder?
29. Two blocks are moving in same direction such that velocity of their center of mass is 10 m/s. What is velocity (in m/s) of first block after head on collision if its velocity before collision is 4 m/s and coefficient of restitution is 0.75?
30. An object is projected from height $h = 4$ m at angle 45° with horizontal. It lands at a horizontal distance $R = 4$ m from the point of projection. The velocity (in m/s) of projection should be ($g = 10\text{m/s}^2$)

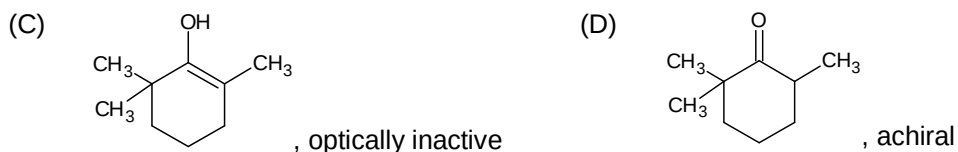
PART – B

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

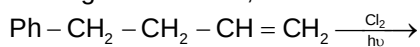
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37. Which of the following IUPAC group of d-block is free from the effect of lanthanide contraction?
 (A) 3
 (B) 4
 (C) 5
 (D) 6
38. Which of the following does not represent appropriate product?
 (A) $\text{CrO}_4^{2-} + \text{H}_2\text{O}_2 \xrightarrow[\text{medium}]{\text{Acidic}} \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$
 (B) $\text{Cr}^{3+} + \text{H}_2\text{O}_2 \xrightarrow[\text{medium}]{\text{Basic}} \text{CrO}_4^{2-} + \text{H}_2\text{O}$
 (C) $\text{MnO}_4^- + \text{H}_2\text{O}_2 \xrightarrow[\text{medium}]{\text{Acidic}} \text{Mn}^{2+} + \text{O}_2$
 (D) $\text{MnO}_4^- + \text{H}_2\text{O}_2 \xrightarrow[\text{medium}]{\text{Basic}} \text{MnO}_2 + \text{O}_2$
39. A metal M does not combine with reagents A and B individually but when combines with an appropriate mixture of reagents A and B produce a gas G_1 . This gas can also be produced by reaction of other metal of same family and dil. solution of reagent. On the basis of above information which of the following statements is/are true?
 (I) These metals may belong to transition metals
 (II) Gas G_1 is paramagnetic and blue color gas
 (III) Metals of same family which are lighter than M produces same gas reaction with reagent B
 (A) II and I
 (B) only I
 (C) only III
 (D) II and III
40. The recommended concentration of fluoride ion in drinking water is up to 1 ppm as fluoride ion is required to make teeth enamel harder by converting $[3\text{Ca}_3(\text{PO}_4)_2 \cdot \text{Ca}(\text{OH})_2]$ to:
 (A) $[3(\text{CaF}_2) \cdot \text{Ca}(\text{OH})_2]$
 (B) $[3\text{Ca}_3(\text{PO}_4)_2 \cdot \text{CaF}_2]$
 (C) $[3\text{Ca}(\text{OH})_2 \cdot \text{CaF}_2]$
 (D) $[\text{CaF}_2]$
41. Photochemical smog occurs in warm, dry and sunny climate:
 (A) reducing smog
 (B) oxidising smog
 (C) both are possible
 (D) none of these

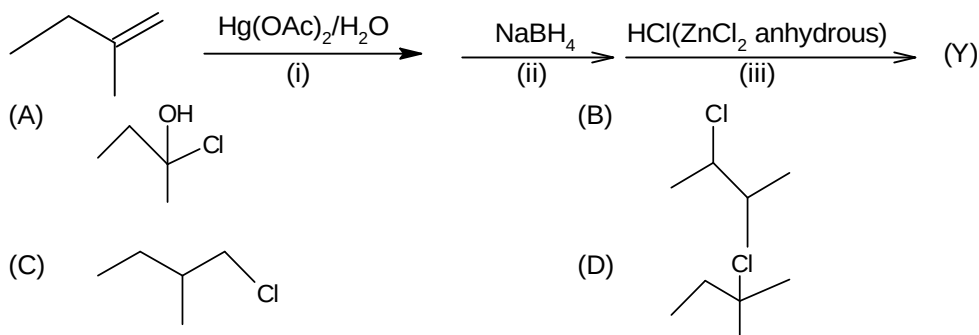




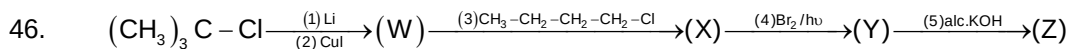
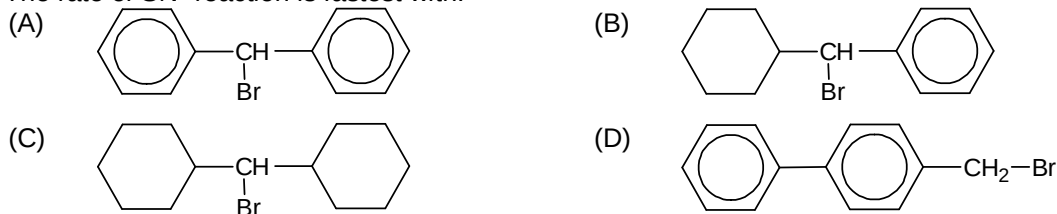
43. In the given reaction, which of the following is the major product?



44.



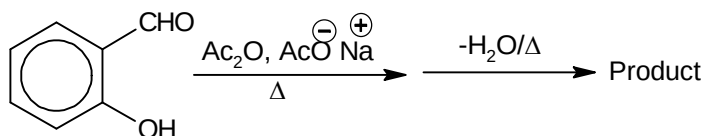
45. The rate of $\text{S}_\text{N}1$ reaction is fastest with:



Which statement is incorrect for above reaction?

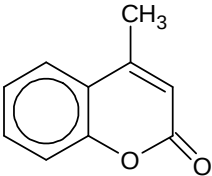
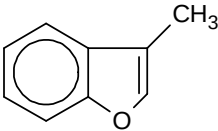
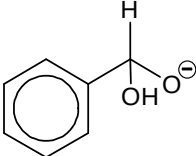
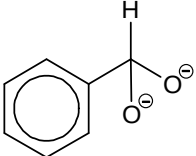
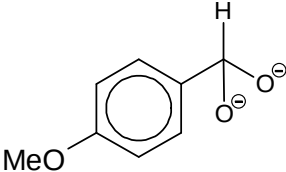
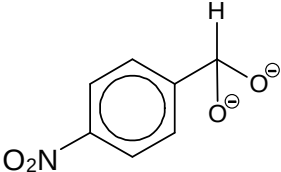
- (A) W is an organometallic compound.
 (B) Formation of X from W takes place via $\text{S}_\text{N}2$ reaction.
 (C) Conversion of X to Y takes place via free radical reaction.
 (D) Conversion of Z from Y take place via E_1 reaction.

47.



Product will be

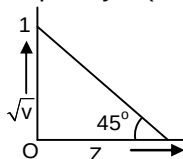


- (C) 
- (D) 
48. In the Cannizzaro's reaction the intermediate that will be the best hydride donor?
- (A) 
- (B) 
- (C) 
- (D) 
49. $C_4H_6O_2$ does not represent:
- (A) A diketone
 (B) A compound with two aldehyde
 (C) An alkenoic acid
 (D) An alkanonic acid
50. The number of isomers of $C_3H_5Br_3$ is (including stereoisomers)
- (A) 4
 (B) 5
 (C) 6
 (D) 7

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. According to Moseley's equation: $\sqrt{\nu} = a(Z - b)$, the graph between $\sqrt{\nu}$ vs Z is given below. The frequency ν (in sec^{-1}) for atomic number (Z) = 52 is:



52. Sum of the two quantum number ($n + l$) for unpaired electrons in third excited state of chlorine ($_{17}\text{Cl}$) is

53. Two solids A and C shows the following dissociation equilibria:
 $A(s) \rightleftharpoons B(g) + D(g)$
 $C(s) \rightleftharpoons E(g) + D(g)$
The pressure over excess solid A is 40 atm and over excess C is 80 atm. The total pressure (in atm) of the gaseous mixture over the solid mixture is:
54. The enthalpy change for first proton neutralization of H_2S is $-33.7 \text{ kJ mol}^{-1}$. What is enthalpy change (in kJ) for first ionization of H_2S ?
55. If pK_b for $C_6H_5O^-$ is 4, what is the pH of 0.1M $C_6H_5O^-$ in solution at $25^\circ C$?
56. Enthalpy of vaporization of a certain liquid is $24.914 \text{ kJ mol}^{-1}$ and the entropy of vaporization is $94.684 \text{ JK}^{-1} \text{ mol}^{-1}$. The temperature at which both the phases coexist is _____ K.
57. The molar heat capacity (in $\text{JK}^{-1} \text{ mol}^{-1}$) of a diatomic gas at constant pressure is:
58. A solid compound AB has NaCl structure. If the radius of the cation is 100 pm (picometer) what is the radius (in pm) of cation B? $1 \text{ pm} = 10^{-12} \text{ m}$.
59. 0.147 g of an organic compound on estimation by Carius method gave 0.287 g of AgCl. What is the percentage of chlorine in organic compound?
60. A sample of 0.50 g of an organic compound was estimated by Kjeldahl's method. The ammonia evolved was absorbed in 50 mL of 0.5 M H_2SO_4 . The residual acid required 60 mL of 0.5 M NaOH for neutralization. Find out the % of nitrogen in organic compound.

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Sum of number of terms in the expansion of $(a_1 + a_2)^{50}, (a_1 + a_2 + a_3)^{50}, (a_1 + a_2 + a_3 + a_4)^{50}, \dots, (a_1 + a_2 + \dots + a_{50})^{50}$ is equal to
- (A) $^{100}C_{50}$
 (B) $^{100}C_{51} - 1$
 (C) $^{100}C_{49}$
 (D) $^{100}C_{50} - 1$
62. Let $A(z_1), B(z_2)$ and $C(z_3)$ be the vertices of an equilateral triangle in the argand plane, such that $|z_1| = |z_2| = |z_3|$, then $\frac{z_2 - z_3}{2z_1 - z_2 - z_3}$
- (A) is purely imaginary
 (B) is real
 (C) is zero
 (D) is of the form $a + ib, ab \neq 0, a, b \in \mathbb{R}$.
63. Let $f(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$, then
- (A) f is an even function if functions of any two rows are all even functions and functions of the remaining row are odd functions.
 (B) f is an odd function if functions of any two rows are all odd functions and functions of the remaining row are even functions.
 (C) f is periodic of period T , if $f_1(x), g_2(x)$ and $h_3(x)$ are periodic of period T .
 (D) none of these
64. A ray of light incident at the point $(-2, -1)$ gets reflected from the tangent at $(0, -1)$ to the circle $x^2 + y^2 = 1$ and the reflected ray touches the circle. The equation of line along which the reflected ray moves is
- (A) $4x - 3y + 5 = 0$
 (B) $4x + 3y + 11 = 0$
 (C) $3x + 4y + 5 = 0$
 (D) $3x - 4y + 2 = 0$

65. If $\int \frac{x + (\sin^{-1} 2x)^3}{\sqrt{1-4x^2}} dx = A\sqrt{1-4x^2} + B(\sin^{-1} 2x)^4 + k$, then values of A and B are
- (A) $A = -\frac{1}{8}, B = \frac{1}{4}$
- (B) $A = -\frac{1}{4}, B = \frac{1}{8}$
- (C) $A = \frac{1}{4}, B = -\frac{1}{8}$
- (D) $A = -\frac{1}{4}, B = -\frac{1}{8}$
66. If the chords of contact of tangents from two points $(-4, 2)$ and $(2, 1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are at right angles then eccentricity of the hyperbola is
- (A) $\sqrt{2}$
- (B) $\sqrt{\frac{3}{2}}$
- (C) $\sqrt{\frac{5}{2}}$
- (D) $\sqrt{3}$
67. The slope of tangent to a curve $y = f(x)$ at $(x, f(x))$ is $1 - 2x$ and the curve passes through the point $(2, -2)$. If the area bounded by the curve and the line $y = \alpha x$ is $\frac{32}{3}$ sq. units. Then the value of α is
- (A) 3
- (B) -3
- (C) 5
- (D) none of these
68. The number of value(s) of $a \in R$ for which the equation $5 \tan^{-1}(x^2 + x + a) + 3 \cot^{-1}(x^2 + x + a) = 2\pi$ has unique solution, is
- (A) 0
- (B) 1
- (C) 2
- (D) infinity
69. If $f(x) > g(x) \forall x \in (a, b)$, where f and g are continuous functions in $[a, b]$, then
- (A) $f'(x) > g'(x) \forall x \in (a, b)$
- (B) $\int_a^b f(x) dx > \int_a^b g(x) dx$
- (C) $f^2(x) > g^2(x), \forall x \in (a, b)$
- (D) none of these

70. If $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the vector $\vec{c}$ satisfying the conditions that (i) it is coplanar with $\vec{a}$ and $\vec{b}$ (ii) it is perpendicular to $\vec{b}$ and (iii) $\vec{a} \cdot \vec{c} = 7$ is
- (A) $-\frac{3}{2}\hat{i} + \frac{5}{2}\hat{j} + 3\hat{k}$
 (B) $-3\hat{i} + 5\hat{j} + 6\hat{k}$
 (C) $-6\hat{i} + 0\hat{j} + \hat{k}$
 (D) $-\hat{i} + 2\hat{j} + 2\hat{k}$
71. A tournament is played on knockout basis. There are n players. In any round if odd number of players are left, one of them is given direct entry into the next round and rest of them play matches. The number of matches played in the tournament, is
- (A) n
 (B) $n - 1$
 (C) $n(n - 1)$
 (D) $\frac{n(n - 1)}{2}$
72. Number of point of discontinuity of the function $f(x) = \begin{cases} \sec^{-1} x, & |x| \geq 1 \\ |x| - 1, & |x| < 1 \end{cases}$ is
- (A) 0
 (B) 1
 (C) 2
 (D) ∞
73. Let $f(x) = \begin{cases} (3 - \sin(1/x))|x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then at $x = 0$ f has a
- (A) maxima
 (B) minima
 (C) neither maxima nor minima
 (D) point of discontinuity
74. Let a, b, c be the sides of a triangle satisfying $\begin{vmatrix} b+c-2a & c+a-2b & a+b-2c \\ bc & ca & ab \\ a^2 & b^2 & c^2 \end{vmatrix} = 0$, then the triangle has to be
- (A) right angled
 (B) isosceles
 (C) equilateral
 (D) isosceles right angled

75. Unit vector $\vec{c}$ is inclined at an angle θ to unit vectors $\vec{a}$ and $\vec{b}$ which are perpendicular. If $c = \lambda(\vec{a} + \vec{b}) + \mu(\vec{a} \times \vec{b})$ where λ, μ be real, then θ belongs to
- (A) $\left(-\frac{\pi}{4}, 0\right)$
- (B) $\left[0, \frac{\pi}{4}\right)$
- (C) $\left[\frac{3\pi}{4}, \pi\right)$
- (D) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$
76. The image of the plane $x + 2y + 3z + a = 0$ in the plane $2x + 4y + 6z + b = 0$ is $3x + 6y + 9z + c = 0$, then
- (A) a, b, c are in A.P.
- (B) $6a, 3b, 2c$ are in A.P.
- (C) a, b, c are in H.P.
- (D) $6a, 3b, 2c$ are in H.P.
77. Solution of differential equation $(x \cos x - \sin x)dx = \frac{x}{y} \sin x dy$ is
- (A) $\sin x = \ln(xy) + C$
- (B) $\ln\left(\frac{\sin x}{x}\right) = y + C$
- (C) $\frac{\sin x}{xy} = C$
- (D) none of these
78. If the line $ax + by + c = 0$, where $a, b, c \in \mathbb{R}$ and $a, b, c \neq 0$, is a normal to the curve $y = x^2$, then $x^3 + 2a^2x + 4a^2c = 0$ has
- (A) three distinct real roots one of which is b
- (B) exactly one real root which is b
- (C) one root is b and two other repeated roots
- (D) none of the roots are equal to b
79. If the mean deviation of number $1, 1+d, 1+2d, \dots, 1+100d$ from their mean is 255, then d is equal to
- (A) 10.0
- (B) 20.0
- (C) 10.1
- (D) 20.2

80. Consider the following statements
 P : Suman is brilliant.
 Q : suman is rich
 R : Suman is honest
 The negation of the statements. "Suman is brilliant and dishonest, if and only if Suman is rich" can be expressed as
- (A) $\sim [Q \leftrightarrow (P \wedge \sim R)]$
 (B) $\sim Q \leftrightarrow (P \wedge R)$
 (C) $\sim Q (P \wedge \sim R) \leftrightarrow Q$
 (D) $\sim P \wedge (Q \leftrightarrow \sim R)$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If $|A| = 5$ and $A = [a_{ij}]_{3 \times 3}$, then $|4A|$ is
82. The number of value(s) of x , which satisfy the equation $\log_{10}(2^{x+1} + x - 37) = x(1 - \log_{10}5)$ is
83. If $f(x^2 - 6x + 6) + f(x^2 - 4x + 4) = 2x \quad \forall x \in R$ then $f(-3) + f(9)$ is
84. If E is an event in a sample space such that $a\sqrt{P(E)} + b\sqrt{P(\bar{E})} = 1$, then the least value of $a^2 + b^2$ is
85. It $\det A = 7$, where $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & b & c \end{bmatrix}$, if $\det(2A)^{-1}$ is m , then find value of $3/m$.
86. 7 white balls and 3 black balls are placed in a row at random. Then the probability that no two black balls are adjacent is P , then value of $45P$.
87. If the integers α and β are chosen at random from 1 to 100, then the probability that $7^\alpha + 7^\beta$ is divisible by 5 is P , then value of $2424P$ is,
88. If 1 lies between the roots of the equation $y^2 - my + 1 = 0$ and $[\cdot]$ denotes the greatest integer function, then $\int_0^{10} \left[\left(\frac{4|x|}{x^2 + 16} \right)^m \right] dx$ is equal to
89. The coefficient of x^{203} in the expression $(x-1)(x^2-2)(x^3-3)\dots(x^{20}-20)$ must be
90. The number of solutions of $\cos x - \sin x = \operatorname{cosec} x$ in $[0, 2\pi]$ are